

The Interactive Effects of Negative Emotion and Reward Motivation on Visual Perception

Sagarika Jaiswal, Lakshman N. C. Chakravarthula, and Srikanth Padmala

Abstract

■ Although there is a rapidly growing interest in reward–emotion interactions, our current understanding of how negative emotion influences reward motivation and modulates reward-driven enhancements in visual perception remains limited. To address these gaps, we conducted an fMRI study using a novel variant of the monetary incentive delay task where the valence (negative or neutral) of an emotional scene image served as a cue to indicate a reward or no-reward prospect in the subsequent house–building discrimination task. During the initial cue stage, we hypothesized competitive interactions between reward anticipation and negative emotion along the common value/valence dimension. However, we instead found independent neural signatures of reward (vs. no-reward) anticipation in the ventral striatum and negative (vs. neutral) emotion in the ventromedial pFC and amygdala, with

a lack of evidence for their interaction. Notably, during the subsequent task stage, we detected an Emotion $\times$ Reward interaction in the parahippocampal gyrus (PHG), wherein reward-driven enhancements in task-related processing were attenuated in the case of negative (vs. neutral) cue images. Furthermore, the Emotion $\times$ Reward interaction scores in PHG and behavioral RTs were correlated across participants. Finally, a regression analysis revealed that negative valence-related activity in ventromedial pFC moderated the relationship between ventral striatum reward anticipation activity and PHG task-related processing. These findings demonstrate that negative emotion and reward motivation, which were largely segregated during the cue stage, interactively modulated subsequent visual perception, thus potentially influencing behavior. ■

INTRODUCTION

Reward motivation and emotion are strong modulators of human behavior. Although investigations of reward motivation and emotion have proceeded largely independently, they are closely related constructs (Sander & Nummenmaa, 2021; Cromwell et al., 2020; Chiew & Braver, 2011; Pessoa, 2009). For instance, the fundamental affective dimensions of value/valence and salience/arousal might be shared between the two (Sander & Nummenmaa, 2021; Park, Kostandyan, Boehler, & Krebs, 2018). Hence, in the past decade, there has been a rapidly growing interest in investigating the interactions between reward motivation and emotion (Padmala, Sambuco, & Pessoa, 2019). One line of previous work investigated how reward motivation (typically manipulated via performance-based monetary rewards) influenced the behavioral and neural correlates of emotional perception and distraction (Chakravarthula & Padmala, 2022; Walsh, Carmel, Harper, & Grimshaw, 2018; Wei, Wang, & Ji, 2016; Choi, Padmala, Spechler, & Pessoa, 2014; Wei & Kang, 2014). However, in a complementary manner, emotional information could also influence the neural signatures of reward motivation, thus modulating associated behaviors,

especially when it competes with the reward value (Yeung et al., 2024; Choi, Padmala, & Pessoa, 2015; Choi et al., 2014). For instance, graphic warnings on cigarette packages might evoke negative emotion, which could potentially dampen the value associated with smoking, thus increasing the intentions to quit (Noar et al., 2016; Kees, Burton, Andrews, & Kozup, 2010). Therefore, it is essential to investigate the neural mechanisms underlying the impact of negative emotion on reward motivation, which is of both basic and clinical relevance.

By employing variants of the classic monetary-incentive delay (MID) task (Knutson, Westdorp, Kaiser, & Hommer, 2000), a limited set of fMRI studies have examined the impact of negative emotion on the neural signatures of reward anticipation. In an initial study, Wittman and colleagues examined how concurrently presented emotional scene images influenced the processing of reward signaling cues (Wittmann, Schiltz, Boehler, & Düzel, 2008). Contrary to the hypothesis, the anticipatory reward (vs. no-reward) responses in multiple brain regions including the dopaminergic midbrain, dorsal striatum, ventral striatum (VS), and ACC were unaffected by the negative (relative to neutral) images. As the authors noted, one potential reason contributing to these null effects might be because the processing of emotional information was implicit or incidental. This was indeed reflected in the weak engagement

of the amygdala by negative (vs. neutral) images, suggesting that the potency of the emotional manipulation might not have been strong enough to modulate reward anticipation responses.

A later set of fMRI studies examined how a relatively more potent threat-of-shock manipulation impacted neural signatures of reward anticipation. In an initial study (Choi et al., 2014), Choi and colleagues found that reward anticipation responses in several brain regions including the midbrain, ventral and dorsal striatum, thalamus, and dorsal ACC were attenuated under threat-of-shock, suggesting that threat might have counteracted reward anticipation. In contrast, Gorka and colleagues found that anticipatory reward responses in the VS (and lateral extrastriate visual cortex) were enhanced in the threat-of-shock context (Gorka, Fuchs, Grillon, & Ernst, 2018). In addition to these conflicting findings, the uncertain threat-of-shock manipulation engages distinct neural substrates compared to those involving a brief presentation of emotional images (Sambuco, Costa, Lang, & Bradley, 2020). Hence, the nature of their influences on reward anticipation might also differ.

Some of the abovementioned methodological concerns were accounted for in a subsequent fMRI study by Park and colleagues (Park, Kostandyan, Boehler, & Krebs, 2019). The authors employed a novel design where the valence of an emotional facial expression signaled a reward or no reward. Specifically, the task mandated the evaluation of emotional valence to indicate whether a face stimulus was linked to a reward or not, thereby rendering both reward and emotion goal-relevant. Although the authors have found signatures of reward (vs. no-reward) processing in the striatum, insula, and SMA, evidence concerning the impact of negative (relative to neutral) emotion on reward-related processing was lacking. As in Wittmann and colleagues (2008), one potential caveat could be the potency of emotional manipulation. Despite being goal relevant, signatures of emotional (vs. neutral) face processing were not detected in the amygdala and fusiform gyrus, regions centrally involved in processing emotional facial expressions (Sabatinelli et al., 2011; Adolphs, 2010). Overall, because of the caveats concerning the potency and task relevance of emotional information, which might have contributed to the discrepant findings in the limited literature so far, our current understanding of the impact of negative emotion on the behavioral and neural signatures of reward motivation is inconclusive and needs further investigation.

Unlike most fMRI studies of reward anticipation (including the ones discussed above), which were designed based on the standard MID task, Park and colleagues (2019) had deliberately not employed a sequential cue–task paradigm because the authors were primarily interested in reward–emotion interactions that could manifest reactively in the absence of any advanced preparatory cues (also see Hu, Padmala, & Pessoa, 2013). So, the employed task design did not provide an opportunity to examine

how emotional valence impacts neural signatures of reward anticipation in the absence of any potential contamination from concurrent task demands or motor-related activity. Second, by separating the cue and task stages, one could examine the influence of integrated reward–emotion signals at the initial cue stage on perceptual processing during the subsequent task stage. Several lines of past work have reported that reward prospect enhanced the perceptual processing of task-related visual stimuli (Cheng, Rich, & Le Pelley, 2021; Takagaki & Krug, 2020; Rothkirch & Sterzer, 2016; Hickey & Peelen, 2015; Pessoa, 2015). However, it is currently unknown how concurrent negative emotion could potentially affect the neural signatures of such reward-driven enhancements in visual perception.

To address these critical gaps, we conducted an fMRI study by employing a modified version of the classic MID task, which allowed us to estimate the neural responses separately during the preparatory cue and the following task stages. During the initial cue stage, relatively more potent emotional scene images (compared to facial expression images; Thom et al., 2014) were used to indicate a reward or no-reward prospect. During the subsequent task stage, participants performed a house–building discrimination task. Specifically, in one experimental phase, negative images signaled an opportunity to win a performance-based monetary reward during the subsequent perceptual discrimination task, whereas a neutral image signaled no such reward opportunity. This instructed emotion–reward mapping was reversed in the other phase, resulting in a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) within-subject factorial design. Similar to Park and colleagues (2019), our design mandated the evaluation of the valence (negative vs. neutral) of an emotional image to determine whether there is an opportunity to win a bonus reward or not, rendering the emotion goal-relevant.

During the cue stage, we were primarily interested in probing the impact of negative emotion on reward anticipation responses in brain regions, including the midbrain, dorsal striatum, VS, and dorsal ACC. Because of the valence incompatibility between positive reward and negative emotion, we expected a competitive interaction along the common valence/value dimension. In particular, we hypothesized that reward (vs. no-reward) anticipation responses in the VS would be attenuated when signaled by a negative compared to a neutral image. The VS is a central node of the distributed reward circuit involved in driving motivated behaviors (Haber & Knutson, 2009) and has been reported to encode emotional and motivational value signals on a common scale (Sescousse, Li, & Dreher, 2015). Second, because reward and emotion manipulations were intertwined in our paradigm, we expected that the neural signatures of negative emotion would be modulated by the concurrent reward (vs. no-reward) anticipation in the amygdala, ventromedial pFC (vmPFC), insula, and lateral occipital

cortex (Lindquist, Satpute, Wager, Weber, & Barrett, 2016; Sabatinelli et al., 2011).

In addition, our design allowed us to investigate the influence of the integrated reward–emotion cues on the subsequent house-versus-building perceptual discrimination task. During the task stage, we were interested in examining responses under different reward–emotion cue conditions in the higher-order ventral visual cortex, specifically the parahippocampal gyrus (PHG), which is centrally involved in the perception of house and building images (Epstein, Harris, Stanley, & Kanwisher, 1999; Aguirre, Zarahn, & D'Esposito, 1998). As noted above, several past studies have reported that reward expectancy enhanced the perceptual processing of task-related stimuli, thus potentially influencing behavior (Antono, Vakhrushev, & Pooresmaeli, 2022; Engelmann, Damaraju, Padmala, & Pessoa, 2009). In this study, we expected that negative images would modulate such reward-driven enhancements in task-related processing. Specifically, we hypothesized a competitive interaction pattern where reward (vs. no-reward) prospect-driven activity in PHG (and corresponding behavioral RT) would be attenuated when signaled by a negative compared to a neutral cue image. Furthermore, because the only difference between the four experimental conditions while performing the discrimination task was the initial cues, we anticipated that the interaction patterns in PHG (and accompanying task performance) could be explained by activity in reward and emotion-sensitive regions during the preceding cue stage. Such findings would provide evidence consistent with the notion of negative emotion interacting with reward motivation to shape visual perception and associated behavior.

METHODS

Participants

Forty participants were recruited for this fMRI study through poster and email advertisements. The sample size was determined based on previous fMRI studies of emotion and reward motivation interactions (Wittmann et al., 2008, $n = 25$; Park et al., 2019, $n = 30$). The participation criteria included an age bracket of 18–35 years, with normal or corrected-to-normal vision, right-handedness, no history of neuropsychiatric disorders or any clinically diagnosed cognitive or motor deficits, no history of substance abuse, and no MRI contraindications. The participants gave their written informed consent to participate in the study and were screened for magnetic resonance safety before the start of the experimental session. The participants were given monetary compensation of Rs. 300 (base pay) for their participation. All experimental procedures were approved by the Institutional Human Ethics Committee of the Indian Institute of Science, Bangalore, and the Central Ethics Committee of HCG Hospital, Bangalore. Data from three participants were excluded: Two of them

exhibited poor task performance (below-chance-level accuracy in one or more experimental conditions), and one other participant had excessive head motion (>6 mm). Data from the remaining 37 (16 women; age: 22.54 ± 3.76 years) participants have been included in the final analyses.

Task Procedures

During the fMRI session, participants rested in a supine position on the patient table and wore noise-attenuating earplugs. They were asked to keep their head still inside the scanner, and some additional foam padding was provided around the head to limit head movement. The experimental stimuli were projected on a screen placed near the foot-end of the table, and participants viewed the screen via a mirror attached to their head coil. Participant responses were collected using the MRI-compatible ResponseGrip acquisition device (NordicNeuroLab). The stimulus presentation and response collection scripts were written in MATLAB (Version R2019a) using functions from the Psychophysics toolbox (Brainard, 1997).

The fMRI session included the calibration task and the main reward–emotion task. The primary purpose of the initial calibration task was to estimate the RT threshold for each stimulus condition separately, which was used as a reward criterion for fast response in the subsequent main task involving emotion and reward manipulations (see below). First, we describe the main reward–emotion task, followed by the calibration task.

Main Reward–Emotion Task

At the start of each trial (Figure 1A), a visual scene stimulus ($4^\circ \times 4^\circ$ of subtended visual angle), either of a negative or neutral valence, was presented centrally for 1000 msec. The valence of the image served as a cue to indicate to participants whether they could win a bonus reward or not on that trial (Cue stage). The cue was followed by a white fixation cross for a variable ISI of 2000–6000 msec. Then, an image of a house or a building ($4^\circ \times 4^\circ$) appeared centrally for 500 msec (Task stage). The participants were instructed to do a house–building categorization task and respond within a 1200-msec response window from the stimulus onset using the top and bottom buttons on the MRI-compatible response grip held in their right hand. If they saw an image of a house, they pressed the top button using their thumb, and for an image of a building, they pressed the bottom one with their index finger (the button-mapping was counterbalanced across participants). Finally, each trial ended with a blank screen of variable intertrial interval (ITI) of 2000–6000 msec. We employed jittered ISI and ITI values to reliably estimate the responses separately during the cue and task stages (Serences, 2004).

The main experimental task was subdivided into four runs: In two runs (say, Phase A; see Figure 1B), a negative image cue signaled reward opportunity, and a neutral

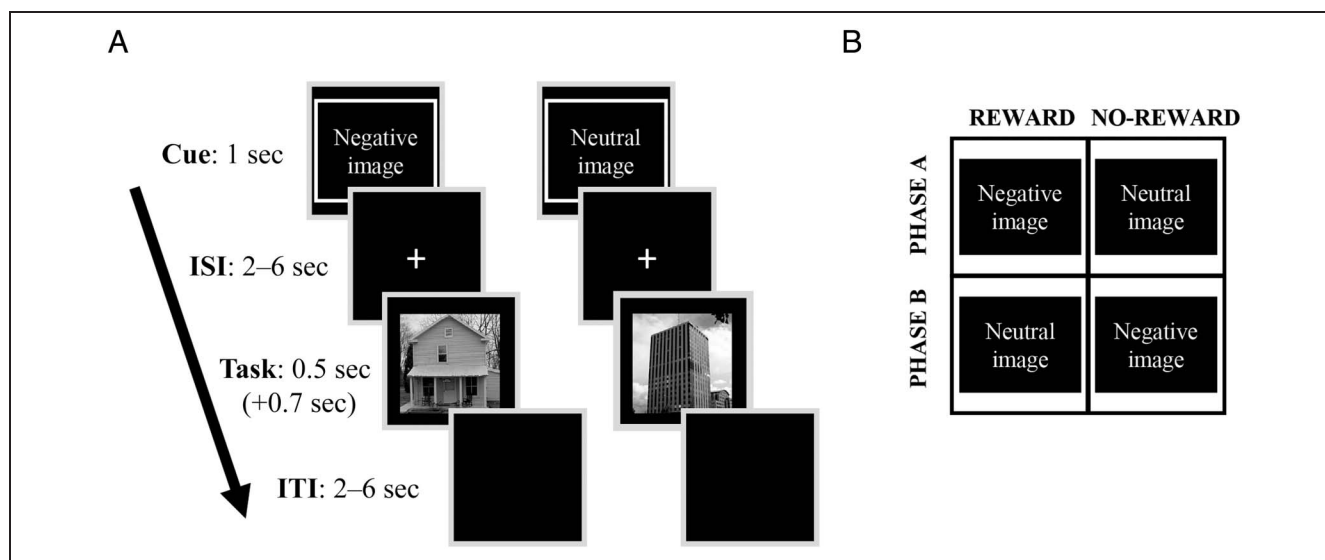


Figure 1. fMRI task paradigm. (A) At the start of each trial, a negative or neutral emotional scene stimulus was presented for 1 sec (cue stage). The valence of the emotional image indicated a performance-based reward or no-reward prospect. After a jittered ISI of 2–6 sec, an image of a house or building was shown for 0.5 sec (task stage). The participants had to indicate if they saw a house or a building within a 1.2-sec response window from the onset of the stimulus (indicated by +0.7 sec). Finally, each trial ended with a variable ITI of 2–6 sec where the screen remained blank. During reward prospect trials, participants received a bonus monetary reward for every correct and fast response. There was no opportunity to win a bonus reward during no-reward trials. (B) The main experimental task was subdivided into four runs: In two runs (say, Phase A), a negative image signaled reward opportunity and a neutral image indicated no reward. In the remaining two runs (say, Phase B), this emotion–reward mapping was reversed. The four runs appeared in an alternating sequence of either ABAB or BABA, and the two types of sequences were counterbalanced across participants.

image cue indicated no reward. In the remaining two runs (say, Phase B), this emotion–reward mapping was reversed. The four runs appeared in an alternating sequence of either ABAB or BABA, and the two types of sequences were counterbalanced across participants. Before the start of each run, participants were explicitly instructed about the emotion–reward mapping for that particular run. On reward prospect trials (signaled by a specific valence in a run), participants earned Rs. 2 for each accurate and fast response, whereas correct but slow responses and incorrect/miss responses yielded no bonus reward. For a response to be considered as fast, participants had to respond earlier than the RT threshold, which was set separately for “house” and “building” image categories based on performance during the initial calibration task (see Calibration Task section below). On no-reward trials, participants had no opportunity to win any bonus reward irrespective of their performance. Participants did not receive trial-by-trial response feedback in the main experimental task. However, at the end of each run, visual text feedback regarding the amount of money earned in that run and the total amount won thus far was displayed. Over the entire experiment, participants won a bonus reward of Rs. 93.9 ± 20.85 , paid at the end of the fMRI session, along with the base pay.

Overall, we employed a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) within-subject factorial design. For each main experimental run of 36 trials, separate pseudorandom trial orders were generated such that the same emotion at the cue stage or the same stimulus category at the task stage were repeated not more than

thrice consecutively. The predetermined trial order of runs was maintained the same across participants. Across the four main experimental runs, there were 144 trials, yielding 36 trials per experimental condition. The variable ISI and ITI values in each run were sampled from an exponential distribution favoring shorter values. In particular, across four runs, the same 36 ISI values were used once in each of the four experimental conditions (in a pseudo-random order) to control for any potential temporal unpredictability effects (Koppe et al., 2014). Finally, at the end of each run, participants were asked to rate their subjective motivational experience on a Likert scale of 1 (*very low*) to 9 (*very high*). The two rating questions asked after Phase A runs were as follows: (1) “How motivated were you to perform the house/building discrimination task after NEGATIVE image signaled REWARD?” and (2) “How motivated were you to perform the house/building discrimination task after NEUTRAL image signaled NO REWARD?” The rating questions asked after Phase B runs were as follows: (1) “How motivated were you to perform the house/building discrimination task after NEUTRAL image signaled REWARD?” and (2) “How motivated were you to perform the house/building discrimination task after NEGATIVE image signaled NO REWARD?”

Before each main experimental task run, participants underwent a short training run (24 trials before the first two runs and 12 trials before the last two) to facilitate learning of the emotion–reward association for the upcoming run. The trial structure in the training runs was the same as that of the main experimental runs except that additional performance-based text feedback (1000 msec) was

provided on each trial after the response window. During reward prospect trials, the following feedback was provided: “Correct and Fast, Won Rs. 2” for fast and accurate trials; “Correct but Slow, Won Rs. 0” for accurate but slow trials; “Incorrect” for incorrect trials; and “No Response” for trials with no response. During the no-reward trials, accurate responses were followed by the feedback “Correct, Won Rs. 0.” Feedback for incorrect and no-response trials was same as that provided during the reward trials.

Calibration Task

The trial structure during the calibration task was similar to that of the main reward–emotion task runs, except that at the initial cue stage, instead of an emotional image, an uninformative gray square ($4^\circ \times 4^\circ$) image was shown. There were 60 trials in total (30 images each of the house and building category during the task stage). As in the main experimental task runs, no feedback about the performance was provided to the participant after every trial. At the end of the calibration run, separate RT distributions were obtained (after excluding error and posterror trials, if any) for house and building categories, and the 30th percentile value of the respective RT distribution was used as the RT threshold for fast response for the reward prospect trials during the subsequent main reward–emotion task. The calibration task was performed during the anatomical scan (see MRI Data Collection section below) and, importantly, without any information about the opportunity to win a potential bonus reward in the subsequent main task. Finally, before the calibration run, a short training run of 16 trials (~2.5 min) was employed to familiarize the participants with the trial structure, types of images, and button mapping.

Stimuli Material

For the cue stage of the main emotion–reward task, colored images of 72 negative and 72 neutral emotion were collected from different standard databases such as the International Affective Picture System (Lang, Bradley, & Cuthbert, 2008) and the Nencki Affective Picture System (Marchewka, Żurawski, Jednoróg, & Grabowska, 2014). None of these emotional images were repeated to prevent potential emotional habituation and reward history effects. Negative stimuli depicted scenes of humans with gore and mutilation, and neutral ones had people involved in regular mundane activities (e.g., sitting on a bench). In a pilot study, we collected subjective valence and arousal ratings of these 144 images from a separate cohort of 16 healthy adult participants (seven women; 24.13 ± 3.01 years old) from the same community. For each image, participants rated their subjective emotional experience on a modified version of the valence–arousal affect grid ratings scale (Russell, Weiss, & Mendelsohn, 1989), with the negative valence dimension of 1–9 (1 = *unpleasant [low]*, 9 = *unpleasant [high]*) on the horizontal axis and

with the arousal dimension of 1–9 (1 = *not at all intense*, 9 = *very intense*) on the vertical axis. On the basis of these ratings, we subdivided the images from negative and neutral categories into two subsets of 36 stimuli each in such a way that valence and arousal ratings differed between categories in each set but were comparable across sets (Supplemental Table S1). These two nonoverlapping stimulus sets were used in Phase A and Phase B runs of the main experimental task. The assignment of a particular stimulus set to one of the phases was counterbalanced across participants. Finally, a separate set of negative and neutral emotion images were used during the cue stage of the training runs for the main emotion–reward task.

For the task stage of the main emotion–reward task, gray-scaled images of 36 houses and 36 buildings were employed from Padmala and Pessoa (2011). These images were normalized in luminance (to the mean of the stimulus set) using the SHINE toolbox in MATLAB (Willenbockel et al., 2010). The house and building images were repeated once in each participant across emotion conditions. For instance, an image of house succeeding a negative image cue in Phase A appeared after a neutral image cue in Phase B. Finally, a separate set of house/building images was used during the initial calibration task and also during the training run before each main experimental task run.

MRI Data Collection

The MRI data were collected using a 32-channel head coil on the 3-T Siemens Skyra MRI scanner at the MRI facility of the HCG hospital, Bangalore. During functional scans, 41 interleaved slices were acquired per volume using the EPI scan sequence (repetition time = 2500 msec, echo time = 28 msec, slice thickness = 3 mm, in-plane resolution = 3 mm $\times$ 3 mm, voxel size = 3 mm isotropic, flip angle = 79°). The functional slices were positioned at about 20° oblique orientation clockwise relative to the line connecting anterior and posterior commissures to minimize susceptibility artifacts in ROIs such as the vmPFC and amygdala. In each of the four main experimental task runs, 131 functional volumes were acquired, which included a 5-sec blank screen at the start to account for MR equilibration effects and a 10-sec blank screen at the end to capture the hemodynamic response of the last trial. Finally, before the start of the functional EPI scans, a T1-weighted whole-brain structural scan was collected using the magnetization prepared rapid gradient echo sequence (repetition time = 2300 msec, echo time = 1.99 msec, in-plane field of view = 256 mm, voxel size = 1 mm isotropic, T1 = 900 msec, flip angle = 9°).

Behavioral Data Analysis

For the RT analysis, error trials ($11.62 \pm 6.31\%$), no-response trials ($1.43 \pm 3.01\%$), and outlier trials ($0.66 \pm 0.67\%$) with RT values beyond 3 SDs from the condition-specific mean were excluded in each participant. For each

participant, mean RT and accuracy rate were determined as a function of Emotion (negative, neutral) and Reward (reward, no-reward), and separate two-way repeated-measures ANOVAs were conducted in the JASP software (Version 0.17.3; Love et al., 2019).

Subjective Ratings Analysis

In each participant, the subjective motivation ratings collected at the end of each run were averaged across two runs of Phase A and Phase B. We then ran a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA on the mean rating scores in the JASP software (Version 0.17.3; Love et al., 2019).

Relationship between Subjective Ratings and Behavioral RT

We examined if there is any association between subjective ratings and behavioral task performance across participants. First, using Pearson correlation analysis, we tested if the Reward $\times$ Emotion interaction scores [(Reward – No-reward)_{NEUTRAL} – (Reward – No-reward)_{NEGATIVE}] in subjective ratings were correlated with the corresponding interaction scores in the RT data. Then, we tested if the main effect of Reward [(Neutral + Negative)_{REWARD} – (Neutral + Negative)_{NO-REWARD}] and the main effect of Emotion [(Reward + No-reward)_{NEGATIVE} – (Reward + No-reward)_{NEUTRAL}] scores in subjective ratings were correlated with the corresponding main effect scores in the RT data. Before conducting Pearson correlation analysis, we checked for any potential outliers using the projection technique that detects multivariate outliers (Pernet, Wilcox, & Rousselet, 2013), but none was detected.

MRI Preprocessing

First, the raw DICOM files from the structural and functional EPI scans were converted into NIFTI format using the Dimon tool from AFNI (Cox, 1996). Before converting to NIFTI, the first two volumes from each functional scan were excluded to account for MR equilibration effects.

The anatomical images were bias-field corrected using *N4BiasFieldCorrection* from ANTs (Avants et al., 2011). Skull stripping was performed using a consensus-based voting scheme by employing algorithms from the following software packages: (i) *antsBrainExtraction.sh* from ANTs, (ii) ROBEX (Iglesias, Liu, Thompson, & Tu, 2011), (iii) *bet* from FMRIB Software Library (FSL; Jenkinson, Beckmann, Behrens, Woolrich, & Smith, 2012), (iv) *3dSkullStrip* from AFNI (Cox, 1996), (v) *mri_watershed* from FreeSurfer (Fischl, 2012), (vi) *bse* from BrainSuite (Shattuck & Leahy, 2002), and (vii) *spm_segment* from SPM (Ashburner & Friston, 2005). Voxels that were classified as in-brain by at least six out of seven packages were considered to be part of the final skull-stripped anatomical image (see Meyer, Padmala, & Pessoa, 2019, for a similar

strategy). The skull-stripped anatomical image was then rotated to the oblique orientation of the functional data (using *3dWarp* from AFNI) and normalized to the skull-stripped MNI152 template at 1-mm resolution using non-linear registration (using *antsRegistrationSyN.sh* program from ANTs).

The preprocessing of functional data involved the following steps. First, the slice-time correction was performed (using *3dTshift* in AFNI) to align the onset times of every slice in a volume to the first acquisition slice. Then, motion correction was performed (using *3dvolreg* in AFNI) to spatially register all volumes to the first volume, which was closest in time to the high-resolution anatomical image. The first motion-corrected volume was then co-registered with the obliquely oriented anatomical image using boundary-based registration (using the *epi-reg* program from FSL). Then, the motion-corrected data were normalized to the MNI152 template by applying the functional–structural co-registration transform along with the affine and nonlinear transforms from the anatomical normalization in a single step and resampled to 2-mm isotropic voxels (using the *antsApplyTransforms* program from ANTs). Subsequently, the normalized functional data were spatially smoothed using a Gaussian kernel of 6-mm FWHM restricted to gray-matter voxels (using *3dBlurInMask* in AFNI). Finally, the mean intensity in each voxel (per run) was scaled to 100 so that the regression estimates from the individual-level GLM analysis (see below) can be interpreted in terms of percent signal change.

Whole-brain Voxel-wise Statistical Analysis

For the individual-level analysis, the representative time series data in each voxel were analyzed in a general linear model (GLM) framework with the following regressors (using *3dDeconvolve* in AFNI). During the cue stage, four regressors were included corresponding to each experimental condition: neutral–no reward, neutral–reward, negative–no reward, and negative–reward. During the task stage, four separate regressors were included corresponding to each of the four cue conditions. As described above, we employed a variable delay between the cue and task stages on each trial. Hence, it is conceivable that anticipatory reward responses could be sustained during the delay period (Padmala, Sirbu, & Pessoa, 2017) and might contaminate the response estimates of subsequent task-stage regressors if not properly accounted for in the GLM. So, to estimate cue-related responses, no assumptions were made about the shape of the hemodynamic response. Responses were estimated from cue stimulus onset to 15 sec using the cubic spline basis function (*CSPLINzero*). As an index of response strength, we used the estimated response at 5 sec after cue onset because of minimal overlap from the succeeding task-stage responses (see Yee, Crawford, Lamichhane, & Braver, 2021, for a similar strategy). To estimate task-stage responses, which were expected to be transient, the four separate task-stage

regressors were modeled for a 0.5-sec stimulus duration from the corresponding stimulus onset and convolved with a canonical gamma variate hemodynamic response function model (Cohen, 1997). The cue and task stage regressors corresponding to no response, error, and outlier trials, whenever applicable, were also included in the model as regressors of no interest (pooled over all conditions). Additional regressors included in the GLM were six estimated motion parameters, their derivatives, and polynomial regressors (of degrees 0–3) separately for each run to account for baseline and drifts of the MR signal. Finally, we excluded volumes with a frame-to-frame displacement of more than 1.5-mm Euclidean distance (half of the original voxel size; Siegel et al., 2014) from the GLM analysis (using the censor option in *3dDeconvolve*).

For the group-level whole-brain voxel-wise analysis, we conducted 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA separately for the cue and task stages using the corresponding estimated beta coefficients (using *3dANOVA3* in AFNI). To account for multiple comparisons, we thresholded the main effects and interaction statistical maps at a voxel-level uncorrected p value of .001 and a minimum cluster extent of 39 voxels for a cluster-level corrected alpha of .05. The cluster-level threshold was estimated by running 100,000 simulations using the *3dClustSim* program in AFNI restricted to gray-matter voxels. For these simulations, the noise smoothness values in three directions were estimated from the GLM residual time series in each participant (using the *3dFWHMx* program in AFNI with the recently added “acf” option) and averaged across participants.

In addition, we ran a whole-brain voxel-wise conjunction analysis (Nichols, Brett, Andersson, Wager, & Poline, 2005), separately for the cue and task stages, to formally test for the presence of both negative emotion and reward anticipation effects in a brain region. First, we created two statistical brain masks based on voxels that showed a significant main effect of the Emotion factor (at cluster-level corrected alpha of .05) and a significant main effect of the Reward factor (at cluster-level corrected alpha of .05). We then created an intersection map of these two masks, which revealed brain regions (a minimum cluster extent of 39 voxels for a cluster-level corrected alpha of .05) commonly activated by the orthogonal emotional and reward manipulations.

ROI Analysis (Cue Stage)

To enhance statistical power, we additionally conducted a priori ROI analysis in the VS, amygdala, and vmPFC. The VS is a key node of the distributed reward anticipation circuit and has been consistently reported in the meta-analyses of MID task fMRI studies (Chen, Chaudhary, & Li, 2022; Jauhar et al., 2021; Oldham et al., 2018; Wilson et al., 2018). On the other hand, amygdala and vmPFC play a central role in emotion processing as supported by the

findings from the meta-analyses of emotion fMRI studies (Yang, Tsai, & Li, 2020; García-García et al., 2016; Lindquist et al., 2016; Sabatinelli, Bradley, Fitzsimmons, & Lang, 2005).

The bilateral amygdala and bilateral VS ROIs were defined using their anatomical definitions based on 50% probabilistic masks from the Harvard-Oxford subcortical structural atlas distributed in FSL (<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/Atlases>). Because vmPFC does not refer to a specific anatomical brain region, we defined vmPFC functional ROI as a 6-mm-radius sphere centered on the peak meta-analytic activation coordinates of encoding emotional valence (MNI coordinates: $x = -9$, $y = -39$, $z = -9$) reported in Lindquist and colleagues (2016). The three ROIs were subsequently downsampled to 2-mm resolution of the preprocessed functional data (using *3dFractionize*). Finally, in each participant, a representative time series for each of the three ROIs was created by averaging the preprocessed time series of all the gray matter voxels inside the ROI (using *3dROIStats*).

For the individual-level GLM analysis, the representative time series data from each ROI were modeled in the same way as described in the Whole-brain Voxel-wise Statistical Analysis section above. For the group-level analysis, our primary focus in the VS, vmPFC, and amygdala ROIs was to examine the interactions between negative emotion and reward anticipation at the cue stage. Hence, for each of these ROIs, estimated beta coefficients of the four cue regressors were submitted to a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA using the JASP software (Love et al., 2019). Accounting for multiple comparisons using the Bonferroni correction for the number of ROIs tested, the p value was set at .016 for these analyses for an overall alpha value of .05.

ROI Analysis (Task Stage)

During the task stage, we primarily focused on the ROI analysis in the PHG, a higher-level visual region involved in the perception of place-related stimuli, including images of houses/buildings (Epstein et al., 1999; Aguirre et al., 1998; Epstein & Kanwisher, 1998). Because we did not include a functional localizer in our scanning protocol because of time constraints, we defined bilateral PHG ROI functionally in a two-step manner. First, we obtained the parahippocampal place area (PPA) ROI from the probabilistic functional ROI atlas of category-selective regions in human ventral visual cortex (Rosenke, van Hoof, van den Hurk, Grill-Spector, & Goebel, 2021) and downsampled it to 2-mm resolution of our preprocessed functional data (using *3dFractionize*). Because the functional PPA ROI was created using multiple place-related stimuli, we aimed to further restrict it to the subset of voxels that responded strongly to house/building images employed in our paradigm. Hence, to create the final PHG ROI, we intersected the functional PPA ROI with the general task activation map during the task stage, which reflects the activation

to house and building stimuli across the four experimental conditions. It is important to emphasize that the general task activation contrast ([1 1 1 1] corresponding to four task stage regressors) used in the PHG ROI definition is orthogonal to the two main effects and interaction contrasts that were of primary interest in this study (see below), thus minimizing the issues of circularity in ROI analysis (Kriegeskorte, Simmons, Bellgowan, & Baker, 2009). Finally, in each participant, a representative time series was created by averaging the preprocessed time series of all the voxels inside the PHG ROI (using *3dROIstats*).

For the individual-level GLM analysis, the representative time series data from PHG ROI was modeled in the same way as described in the Whole-brain Voxel-wise Statistical Analysis section. Our primary focus in the PHG ROI was to examine how the orthogonal emotion and reward manipulations during the preceding cue stage influenced processing of house/building images at the subsequent task stage. Hence, for PHG ROI, estimated beta coefficients of the four task regressors were submitted to a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA using the JASP software (Love et al., 2019).

Relationship between Task-stage PHG Responses and Behavior

In the context of our paradigm, we expected individual differences in PHG task-related processing might be related to individual differences in behavioral performance during the discrimination task. Specifically, we wanted to examine if the Emotion $\times$ Reward interaction scores in PHG were correlated with the corresponding interaction scores in the behavioral RT data. Hence, across participants, we ran a Pearson correlation analysis between PHG ROI interaction scores [(Reward – No-reward)_{NEUTRAL} – (Reward – No-reward)_{NEGATIVE}] and the corresponding RT interaction scores [(Reward – No-reward)_{NEUTRAL} – (Reward – No-reward)_{NEGATIVE}]. Before conducting Pearson correlation analysis, we checked for any potential outliers using the projection technique that detects multivariate outliers (Pernet et al., 2013), but none was detected.

Relationship between Cue-stage Responses and Task-stage PHG and RT Responses

In our paradigm, the only difference between the four experimental conditions, while participants were performing the house–building categorization task, was the preceding reward–emotion combination cue. So, first, we examined if the individual differences in the negative emotion and reward anticipation responses during the initial cue stage explain the variability in PHG responses during the subsequent task stage. Specifically, we expected that individuals with higher negative valence-related activity

would exhibit greater attenuation of reward-driven enhancements in task-related processing, giving rise to stronger competitive interactions between negative emotion and reward anticipation in PHG. To formally test this hypothesis, we ran a statistical moderation analysis (Baron & Kenny, 1986) using a multiple linear regression model in the JASP software. In the regression model, we employed the main effect of Emotion in vmPFC at the cue stage as an index of negative valence (Park et al., 2018; Lindquist et al., 2016) and the main effect of Reward in the VS as an index for reward anticipation (Chen et al., 2022; Oldham et al., 2018) as well as the interaction between the two as independent variables. The Emotion $\times$ Reward interaction in PHG at the task stage was used as the dependent variable.

The regression model was set up as follows:

$$\begin{aligned} \text{PHG interaction score}_i = & b_0 + b_{\text{VS}} \text{VS rew main eff}_i \\ & + b_{\text{vmPFC}} \text{vmPFC emo main eff}_i \\ & + b_{\text{VS} \times \text{vmPFC}} (\text{VS rew main eff}_i \\ & * \text{vmPFC emo main eff}_i) \end{aligned}$$

For each participant i , the PHG ROI interaction score was calculated as [(Reward – No-reward)_{NEUTRAL} – (Reward – No-reward)_{NEGATIVE}]. The main effect scores for Reward in VS ROI and Emotion in vmPFC ROI were calculated as [(Negative + Neutral)_{REWARD} – (Negative + Neutral)_{NO-REWARD}] and [(Reward + No-reward)_{NEGATIVE} – (Reward + No-reward)_{NEUTRAL}], respectively. The variance inflation factors of the three predictors in the regression model were between 1.3 and 1.7, indicating that the independent variables were minimally correlated, thus ensuring the reliability of the estimated regression coefficients (Kutner, Nachtsheim, Neter, & Li, 2005).

In the regression model described above, we used the main effect of emotion in vmPFC (and not the amygdala) because past work has consistently reported that fMRI responses in vmPFC were sensitive to the valence dimension of emotion (Lindquist et al., 2016; Heller, Lapate, Mayer, & Davidson, 2014; Winecoff et al., 2013). In contrast, univariate fMRI responses in the amygdala were more frequently reported to be sensitive to the arousal (rather than valence) dimension of emotion (Lin et al., 2020; Sabatinelli et al., 2005; Phan et al., 2003; but see Anders, Eippert, Weiskopf, & Veit, 2008). In this study, because we were primarily interested in examining competitive interactions between negative emotion and reward anticipation along the common value/valence dimension, we employed the main effect of emotion in vmPFC as an index of negative valence response.

In a separate regression model, we additionally examined if the individual differences in the negative emotion and reward anticipation responses during the initial cue stage explain the variability in behavioral RT responses during the subsequent task stage.

The regression model was set up as follows:

$$RT \text{ interaction score}_i = b_0 + b_{VS} VS \text{ rew main eff}_i + b_{vmPFC} vmPFC \text{ emo main eff}_i + b_{VS*vmPFC} (VS \text{ rew main eff}_i * vmPFC \text{ emo main eff}_i)$$

For each participant i , the RT interaction score was calculated as $[(\text{Reward} - \text{No-reward})_{\text{NEUTRAL}} - (\text{Reward} - \text{No-reward})_{\text{NEGATIVE}}]$.

Bayesian Analyses

For all the analyses reported in the main article (except the whole-brain voxel-wise analyses), we additionally conducted the corresponding Bayesian analysis in JASP (with default priors) to calculate the inverse Bayes factor (BF_{01}) values to estimate the likelihood of the absence of the effect (i.e., evidence in favor of the null hypothesis; Keyesers, Gazzola, & Wagenmakers, 2020). The BF_{01} values between 1 and 3 indicate “anecdotal” evidence for the

corresponding null hypothesis, and BF_{01} values between 3 and 10 indicate “moderate” evidence for the corresponding null hypothesis (Wagenmakers et al., 2018).

RESULTS

Behavioral Data

In the RT data (Figure 2A, Table 1), the 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA revealed a main effect of Reward, $F(1, 36) = 67.89, p < .001, \eta_p^2 = .48, BF_{01} = 6.59e-8$. Confirming our reward motivation manipulation, responses were faster during reward compared to no-reward trials. However, neither the main effect of Emotion, $F(1, 36) = 0.071, p = .79, \eta_p^2 = 2.05e-4, BF_{01} = 4.86$, nor the Emotion $\times$ Reward interaction, $F(1, 36) = 7.28e-4, p = .979, \eta_p^2 = 3.33e-6, BF_{01} = 3.51$, were detected in the RT data. In the accuracy data (Figure 2B, Table 1), no significant main effect of Reward, $F(1, 36) = 0.066, p = .798, \eta_p^2 = 5.76e-4, BF_{01} = 4.54$, Emotion, $F(1, 36) = 1.08, p = .306, \eta_p^2 = .008, BF_{01} =$

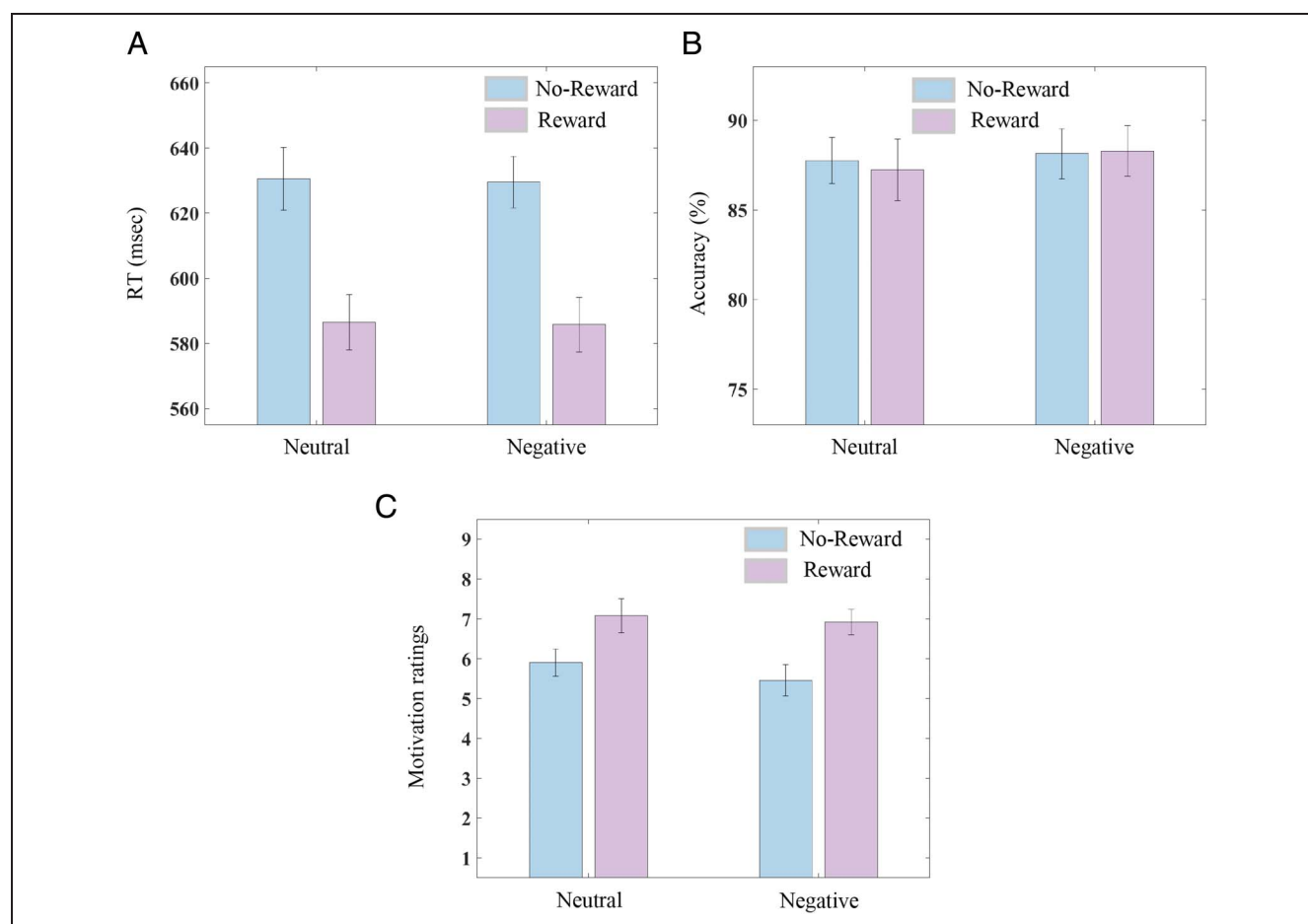


Figure 2. Behavioral and subjective ratings results. (A) RT data revealed a main effect of Reward, whereas the main effect of Emotion and Emotion $\times$ Reward interaction were not detected. (B) Accuracy data did not reveal either of the main effects or interaction. (C) Subjective motivation ratings (1 = *not motivated at all* and 9 = *highly motivated*) collected at the end of each run revealed the main effects of both Reward and Emotion but not the interaction. In each subpanel, the violin plot shows the distribution of corresponding values of the four experimental conditions of interest. The box plots within the violins indicate the interquartile range (first quartile to third quartile), and white dots inside black circles show median values. Each colored dot represents the value from a participant, and each line connects the values of four experimental conditions in a participant.

Table 1. Behavioral Results

Cue Condition	RT	Accuracy (%)
	Mean $\pm$ SD	Mean $\pm$ SD
Neutral–no reward	630.57 $\pm$ 66.46	87.76 $\pm$ 8.16
Neutral–reward	586.57 $\pm$ 69.66	87.24 $\pm$ 6.97
Negative–no reward	629.54 $\pm$ 66.47	88.14 $\pm$ 7.28
Negative–reward	585.78 $\pm$ 68.63	88.29 $\pm$ 7.53

This table shows descriptive RT and accuracy values during the house–building categorization task for the four emotion–reward cue conditions.

3.36), and their interaction, $F(1, 36) = 0.168, p = .684, \eta_p^2 = .002, BF_{01} = 3.39$, were detected.

Subjective Motivation Ratings

The 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA on the subjective motivation ratings (Figure 2C, Table 2) revealed the main effects of both Reward, $F(1, 36) = 23.13, p < .001, \eta_p^2 = .303, BF_{01} = 0.001$, and Emotion, $F(1, 36) = 5.13, p = .030, \eta_p^2 = .016, BF_{01} = 0.74$. The motivation ratings were higher when the cue signaled a reward compared to the no-reward prospect. On the other hand, motivation ratings were lower when the cue image was negative compared to neutral emotion. However, the interaction between Emotion and Reward was not detected, $F(1, 36) = 1.35, p = .254, \eta_p^2 = .004, BF_{01} = 2.03$.

Relationship between Subjective Ratings and Behavioral RT

To examine if there is any association between subjective ratings and behavioral RT, we conducted Pearson correlation analyses across participants. The correlation analyses between the Reward $\times$ Emotion interaction scores in subjective ratings and the corresponding RT interaction scores also did not reveal any significant association, $r(36) = -.24, p = .158, BF_{01} = 1.87$ (Supplemental Figure S1). In addition, the main effect of Reward and

Emotion scores in subjective ratings and the corresponding RT main effect scores were not correlated (Supplemental Figure S1; reward scores: $r(36) = -.23, p = .163, BF_{01} = 1.91$; emotion scores: $r(36) = -.11, p = .510, BF_{01} = 3.96$). Of note, these null results should be interpreted with caution, as unlike the RT data, subjective ratings of the different reward–emotion cue combinations were not collected on each trial during the task.

fMRI Whole-brain Voxel-wise Analysis

Our primary research question in this study was to examine the impact of negative emotion on the neural signatures of reward anticipation. To do so, in a whole-brain voxel-wise analysis, we ran a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA using beta estimates separately from the cue and task stages. During the cue stage, we found several clusters exhibiting a significant main effect of Emotion including the amygdala, fusiform gyrus/inferior temporal gyrus, mid-insula, and intraparietal sulcus (Figure 3, Table 3). In all these clusters, negative images evoked higher activation than neutral ones. We also observed a significant main effect of Reward in multiple regions including the ventral and dorsal striatum, anterior insula, dorsal ACC, FEFs, and intraparietal sulcus (Figure 3, Table 4). In all these clusters, prospect of performance-based reward evoked higher activation than no-reward prospect. Crucially, we did not detect any clusters with significant interaction effects between negative emotion and reward anticipation at the cue stage. Similarly, during the task stage, we found some clusters exhibiting significant main effects of Emotion and Reward (Figure 4, Tables 5–6). Concerning our primary research question, we did not detect any clusters exhibiting interactions between negative emotion and reward anticipation at the task stage.

Next, to identify the common brain regions recruited by both negative emotion and reward anticipation, we ran a whole-brain voxel-wise conjunction analysis based on the significant main effects, separately for the cue and task stages. During the cue stage, the conjunction analysis revealed overlapping main effects of emotion and reward in the right intraparietal sulcus (rIPS) and bilateral lateral

Table 2. Subjective Motivation Ratings: Average Self-reported Motivation Ratings for the Four Emotion–Reward Cue Conditions on a Likert Scale of 1–9 (1 = *Not Motivated at All* and 9 = *Highly Motivated*)

Rating Question	Mean $\pm$ SD
How motivated were you to perform the house/building discrimination task after NEGATIVE image signaled REWARD?	6.92 $\pm$ 1.25
How motivated were you to perform the house/building discrimination task after NEUTRAL image signaled NO REWARD?	5.91 $\pm$ 1.69
How motivated were you to perform the house/building discrimination task after NEUTRAL image signaled REWARD?	7.08 $\pm$ 1.39
How motivated were you to perform the house/building discrimination task after NEGATIVE image signaled NO REWARD?	5.46 $\pm$ 1.91

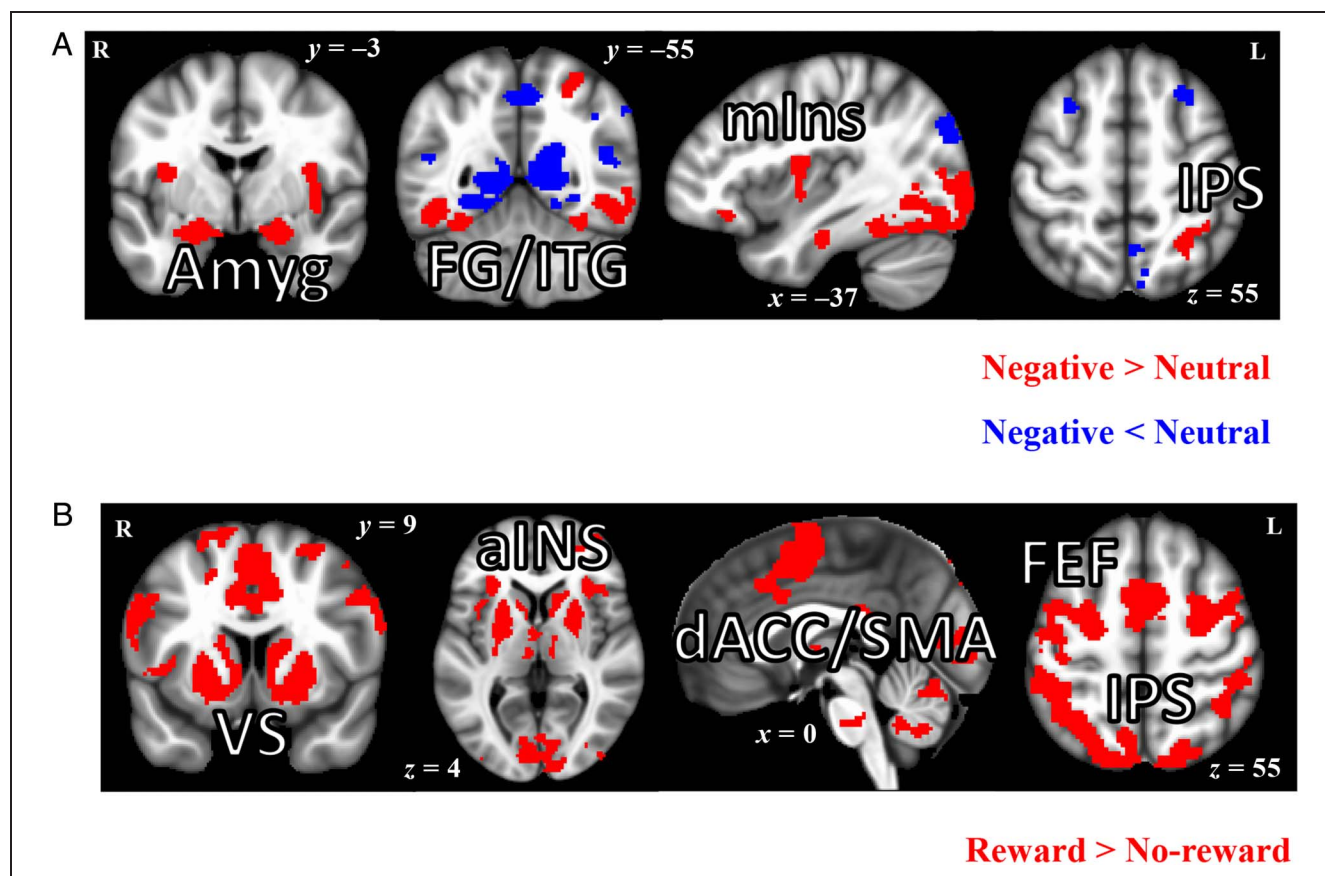


Figure 3. Cue stage whole-brain voxel-wise functional MRI results displayed at an uncorrected $p = .001$ and 39-voxel cluster extent (cluster-level alpha of .05). (A) Clusters exhibiting a significant main effect of Emotion are shown. (B) Clusters exhibiting a significant main effect of Reward are shown. Because cluster-extent based thresholding was used for multiple-comparison correction, voxels are displayed using a binary threshold. For further rationale about using binary maps in the context of cluster-based thresholding, see Woo, Krishnan, and Wager (2014). Amyg = amygdala; FG = fusiform gyrus; ITG = inferior temporal gyrus; mIns = mid-insula; IPS = intraparietal sulcus; aINS = anterior insula; dACC = dorsal ACC; L = left; R = right.

occipital cortex (Figure 5, Table 7). The presence of two main effects with a lack of evidence for significant interaction in these regions suggests that the effects of negative emotion and reward anticipation were additive in nature. During the task stage, no clusters were detected in the conjunction analysis.

fMRI ROI Analysis (Cue Stage)

In addition, to enhance statistical power, we conducted focused ROI analyses in a very limited number of our a priori ROIs. In the VS ROI, we performed a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures ANOVA on the beta estimates of the cue regressors (Figure 6A). We detected a significant main effect of Reward, $F(1, 36) = 16.73$, $p < .001$, $\eta_p^2 = .102$, $BF_{01} = 0.05$, with higher activation during reward (relative to no-reward) cues. However, main effect of Emotion, $F(1, 36) = 0.17$, $p = .684$, $\eta_p^2 = .001$, $BF_{01} = 4.47$, and Emotion $\times$ Reward interaction, $F(1, 36) = 0.09$, $p = .758$, $\eta_p^2 = 9.956e-4$, $BF_{01} = 3.82$, were not detected. In addition, we were interested in examining how concurrent reward anticipation influenced negative emotion

processing in the vmPFC and amygdala. In the vmPFC ROI (Figure 6B), we detected a significant main effect of Emotion, $F(1, 36) = 7.26$, $p < .011$, $\eta_p^2 = .049$, $BF_{01} = 0.64$, where negative images evoked lower responses compared to neutral ones. However, main effect of Reward, $F(1, 36) = 1.21$, $p = .28$, $\eta_p^2 = .008$, $BF_{01} = 3.43$, and Emotion $\times$ Reward interaction, $F(1, 36) = 1.11$, $p = .299$, $\eta_p^2 = .013$, $BF_{01} = 1.82$, were not detected. In the amygdala ROI (Figure 6C), we detected a significant main effect of Emotion, $F(1, 36) = 32.48$, $p < .001$, $\eta_p^2 = .171$, $BF_{01} = 0.001$, where negative images evoked higher responses compared to neutral ones. However, main effect of Reward, $F(1, 36) = 2.01$, $p = .165$, $\eta_p^2 = .01$, $BF_{01} = 3.24$, and Emotion $\times$ Reward interaction, $F(1, 36) = 7.93e-4$, $p = .978$, $\eta_p^2 = 1.004e-5$, $BF_{01} = 3.89$, were not detected. Overall, these results indicate that the neural correlates of reward anticipation and negative emotion at the cue stage were largely segregated in our ROIs, with little evidence for the interaction between the two.

Furthermore, in these ROIs, we also examined the interactions between negative emotion and reward anticipation at the subsequent task stage (see Supplemental Figure S2 for responses and Supplemental Table S2 for

Table 3. Clusters Exhibiting Main Effect of Emotion That Survived Multiple-Comparison Correction in Whole-brain Voxel-wise Analysis at the Cue Stage (Peak MNI Coordinates, Cluster Size [k], $F(1, 36)$ Values, and Unstandardized Effect Size in Terms of Percent Signal Change [ES])

Negative > Neutral

$$ES = [(Reward + No-Reward)_{Negative} - (Reward + No-Reward)_{Neutral}]$$

Cluster	x	y	z	k	$F(1, 36)$	ES
Left inferior temporal gyrus/lateral occipital cortex/parahippocampal gyrus/fusiform gyrus	-54	-72	-12	2617	32.91	0.41
Right inferior temporal gyrus/lateral occipital gyrus/parahippocampal gyrus/fusiform gyrus	58	-66	-12	1738	43.23	0.44
Left amygdala	-8	-2	-18	319	18.73	0.35
Right superior parietal lobule/intraparietal sulcus	38	-44	56	219	25.87	0.13
Left supramarginal gyrus	-64	-24	44	209	27.19	0.17
Right supramarginal gyrus	64	-22	46	190	27.22	0.18
Right amygdala	20	-2	-16	187	39.09	0.20
Left orbitofrontal gyrus	-30	36	-12	180	40.58	0.14
Right insula	42	-2	-2	147	27.42	0.11
Right inferior frontal gyrus	52	44	10	146	37.83	0.16
Left insula	-36	-6	14	138	81.67	0.15
Left inferior temporal gyrus	-34	-24	-22	113	30.00	0.13
Left occipital gyrus	-22	-64	32	109	18.08	0.19
Right posterior angular gyrus	28	-74	36	104	35.17	0.20
Right inferior frontal gyrus	44	6	24	88	16.27	0.14
Right orbitofrontal gyrus	26	28	-14	59	27.87	0.14

Neutral > Negative

$$ES = [(Reward + No-Reward)_{Neutral} - (Reward + No-Reward)_{Negative}]$$

Cluster	x	y	z	k	$F(1, 36)$	ES
Precuneus/parahippocampal gyrus/fusiform gyrus	-28	-48	-6	7239	71.05	0.29
Right posterior angular gyrus/middle temporal gyrus	40	-82	36	818	35.51	0.19
Right middle frontal gyrus	22	26	44	342	20.96	0.07
Left middle frontal gyrus	-44	6	58	231	12.43	0.09
Left posterior angular gyrus	-34	-88	34	230	24.43	0.15
Right angular gyrus	48	-60	54	156	19.72	0.11
Left middle temporal gyrus	-66	-46	12	143	14.48	0.09
Right STS	70	-24	-2	97	13.34	0.11
Right middle frontal gyrus	36	2	64	64	12.92	0.12
Right superior temporal gyrus	68	-10	6	57	12.07	0.11
Right anterior temporal gyrus	44	-28	26	53	12.15	0.08

Table 4. Clusters Exhibiting Main Effect of Reward That Survived Multiple-Comparison Correction in Whole-brain Voxel-wise Analysis at the Cue Stage (Peak MNI Coordinates, Cluster Size [k], $F(1, 36)$ Values, and Unstandardized Effect Size in Terms of Percent Signal Change [ES])*Reward > No Reward*

$$ES = [(Negative + Neutral)_{Reward} - (Negative + Neutral)_{No-Reward}]$$

Cluster	x	y	z	k	$F(1, 36)$	ES
SMA/dorsal ACC/primary motor cortex/superior frontal gyrus	-2	4	56	5423	24.92	0.21
Left parietal cortex	-6	-78	56	3037	34.65	0.25
Left cerebellum (lateral zone)/inferior temporal gyrus/lateral occipital temporal sulcus	-56	-58	-26	1357	18.41	0.26
Left VS/putamen/caudate	-10	-4	14	1162	18.60	0.34
Right VA/putamen/caudate	18	0	-12	1104	20.63	0.17
Calcarine sulcus	6	-100	2	1084	13.04	0.28
Right cerebellum (lateral zone)	20	-86	-18	1078	14.71	0.24
Cerebellum (posterior lobe/intermediate zone/vermis)	30	-56	-60	1067	13.84	0.14
Right parietal lobule/supramarginal gyrus/postcentral sulcus	44	-28	36	733	13.91	0.11
Left middle/inferior frontal gyrus	-38	58	18	451	24.65	0.15
Right middle/inferior frontal gyrus	34	58	22	384	26.65	0.10
Right superior parietal lobule/posterior angular gyrus	6	-78	58	369	15.95	0.30
Left anterior insula	-54	18	-4	284	19.89	0.18
Right anterior insula	36	22	8	217	19.27	0.09
Left thalamus	-4	-2	4	180	14.07	0.21
Right inferior occipital gyrus	44	-88	4	175	17.31	0.12
Left supramarginal gyrus	-58	-32	28	140	16.20	0.08
Right middle temporal gyrus	56	-20	-8	138	21.43	0.09
Left thalamus	-16	-26	-8	107	13.85	0.13
Left middle frontal gyrus	-38	38	40	101	20.74	0.10
Posterior cingulate cortex	0	-34	24	97	15.79	0.16
Right thalamus	12	-16	6	95	15.84	0.09
Right posterior angular gyrus	22	-96	24	87	22.16	0.11
Right middle frontal gyrus	38	42	40	69	29.67	0.10
Right postcentral sulcus/supramarginal gyrus	56	-20	34	61	19.91	0.07
Brainstem	4	-34	-30	60	14.48	0.09
Left inferior occipital gyrus	-36	-94	-12	59	13.74	0.11
Right thalamus	16	-26	-10	50	14.41	0.13
Superior occipital gyrus	-2	-86	42	49	14.47	0.16

No-Reward > Reward

$$ES = [(Negative + Neutral)_{No-Reward} - (Negative + Neutral)_{Reward}]$$

No clusters found

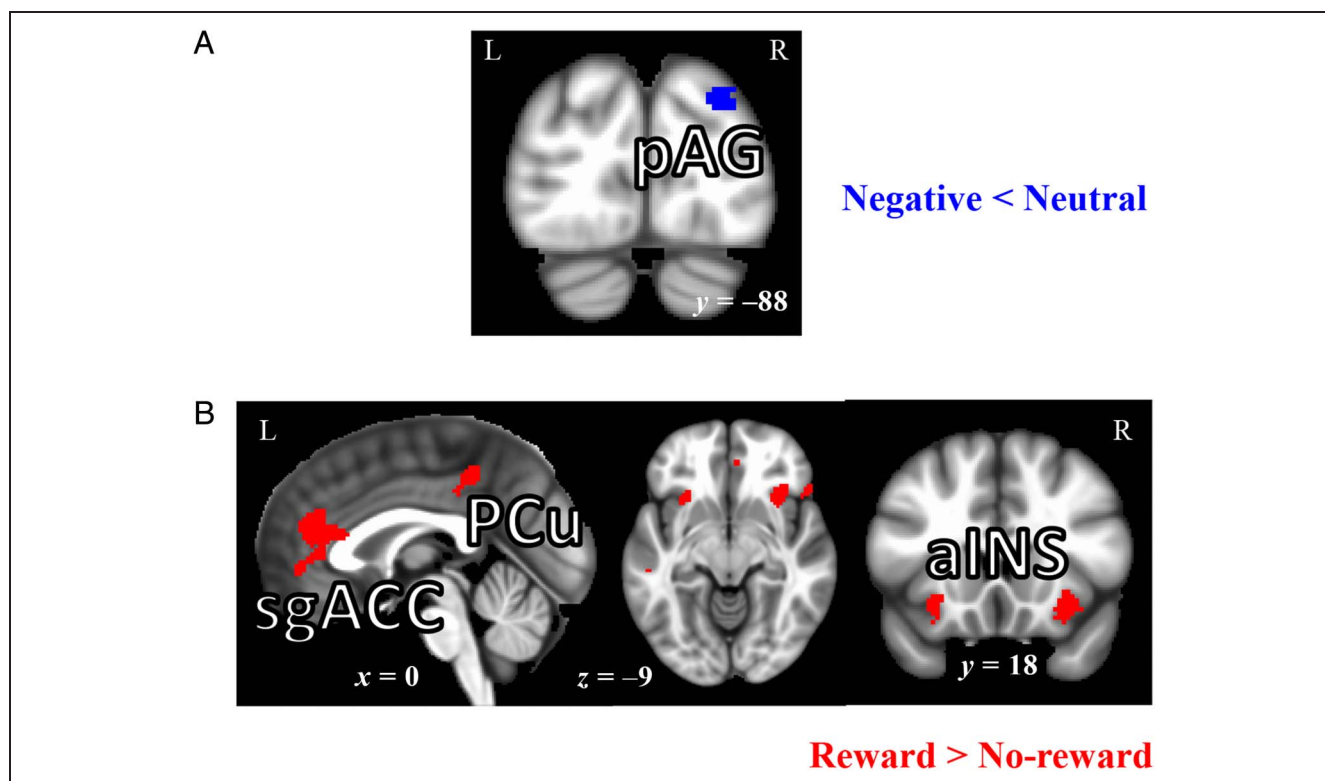


Figure 4. Task stage whole-brain voxel-wise functional MRI results displayed at an uncorrected $p = .001$ and 39-voxel cluster extent (cluster-level alpha of .05). (A) Clusters exhibiting a significant main effect of Emotion are shown. (B) Clusters exhibiting a significant main effect of Reward are shown. Because cluster-extent-based thresholding was used for multiple-comparison correction, voxels are displayed using a binary threshold. For further rationale about using binary maps in the context of cluster-based thresholding, see Woo and colleagues (2014). pAG = posterior angular gyrus; sgACC = subgenual ACC; PCu = precuneus; aINS = anterior insula; L = left; R = right.

results). As in the cue stage, we did not detect any Reward $\times$ Emotion interactions during the task stage.

fMRI ROI Analysis (Task Stage)

During the task stage, we were primarily interested in examining how the different reward–emotion cues influenced subsequent task-related processing in PHG. Accordingly, in the PHG ROI, a 2 Emotion (negative, neutral) $\times$ 2 Reward (reward, no-reward) repeated-measures

ANOVA on the beta estimates of the task regressors (Figure 7A) revealed a main effect of Reward, $F(1, 36) = 4.80, p = .035, \eta_p^2 = .039, BF_{01} = 0.92$, with higher activation during reward (relative to no-reward) trials. The main effect of Emotion was not detected, $F(1, 36) = 2.19, p = .148, \eta_p^2 = .019, BF_{01} = 2.14$. Critically, we detected a significant Emotion $\times$ Reward interaction, $F(1, 36) = 4.68, p = .037, \eta_p^2 = .039, BF_{01} = 0.31$. The follow-up paired t tests (accounting for multiple comparisons using the Bonferroni correction) revealed that the two-way

Table 5. Clusters Exhibiting Main Effect of Emotion That Survived Multiple-Comparison Correction in Whole-brain Voxel-wise Analysis at the Task Stage (Peak MNI Coordinates, Cluster Size [k], $F(1, 36)$ Values, and Unstandardized Effect Size in Terms of Percent Signal Change [ES])

Negative > Neutral

$$ES = [(Reward + No-Reward)_{Negative} - (Reward + No-Reward)_{Neutral}]$$

No clusters found

Negative > Neutral

$$ES = [(Reward + No Reward)_{Neutral} - (Reward + No Reward)_{Negative}]$$

Cluster	x	y	z	k	$F(1, 36)$	ES
Right posterior angular gyrus	30	−88	34	64	12.84	0.21

Table 6. Clusters Exhibiting Main Effect of Reward That Survived Multiple-Comparison Correction in Whole-brain Voxel-wise Analysis at the Task Stage (Peak MNI Coordinates, Cluster Size [k], $F(1, 36)$ Values, and Unstandardized Effect Size in Terms of Percent Signal Change [ES])

Reward > No-Reward

$$ES = [(Negative + Neutral)_{Reward} - (Negative + Neutral)_{No-Reward}]$$

Cluster	x	y	z	k	$F(1, 36)$	ES
Subgenual ACC	0	42	18	575	17.54	0.23
Right anterior insula	54	24	-8	266	18.48	0.33
Superior frontal gyrus (medial part)	2	26	50	164	15.82	0.18
Right supramarginal gyrus	58	-38	52	161	14.41	0.15
Left supramarginal gyrus/angular gyrus	-56	-58	46	143	13.22	0.17
Precuneus/posterior cingulate gyrus	0	-38	46	109	23.50	0.22
Left middle temporal gyrus	-58	-30	-4	103	15.64	0.17
Left lingual gyrus/occipital gyrus/precuneus	-4	-46	4	79	16.32	0.38
Left anterior insula	-32	18	-12	62	14.89	0.15
Right middle frontal gyrus	48	14	52	52	12.85	0.19
Right lingual gyrus	16	-82	-16	40	15.35	0.27

No-Reward > Reward

$$ES = [(Negative + Neutral)_{No-Reward} - (Negative + Neutral)_{Reward}]$$

No clusters found

Figure 5. Cue stage whole-brain voxel-wise conjunction analysis results. Cluster that exhibited conjunction of main effects of Emotion and Reward factors in rIPS. The bar plot shows the average group-level estimates of the four experimental conditions of interest during the cue stage. It should be noted that the primary purpose of these average bar plots is to illustrate the overlapping pattern of main effects and not to draw any statistical inference. Hence, error bars were not plotted on top of the average estimates to avoid issues of circularity (Kriegeskorte et al., 2009). IPS = intraparietal sulcus; L = left; R = right.

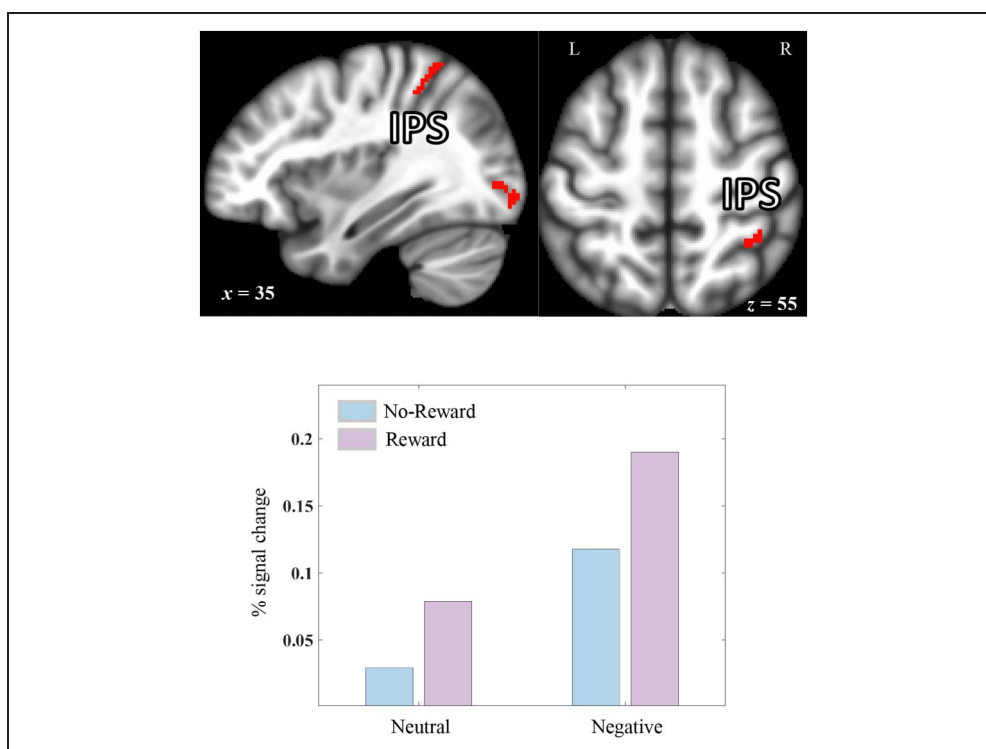


Table 7. Cue Stage Conjunction Analysis of Main Effects of Emotion and Reward

Cluster	<i>x</i>	<i>y</i>	<i>z</i>	<i>k</i>
Calcarine sulcus	−2	−92	−2	400
Right lateral occipital cortex	30	−92	−10	95
Right intraparietal sulcus	36	−42	48	69
Left lateral occipital cortex	−34	−92	−10	45

The following clusters survived multiple-comparison correction in whole-brain voxel-wise conjunction analysis at the cue stage (MNI coordinates and cluster size [*k*]).

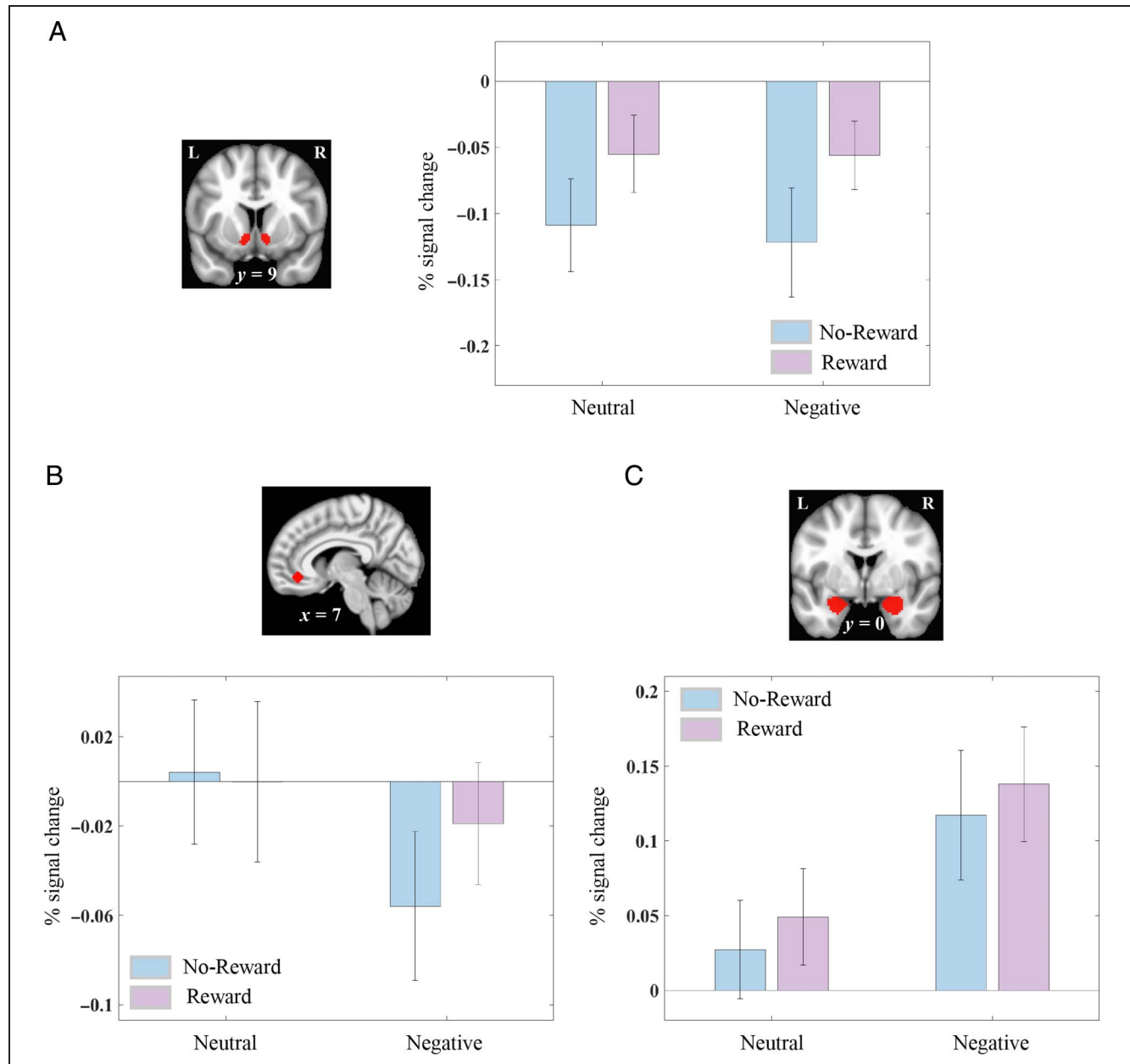
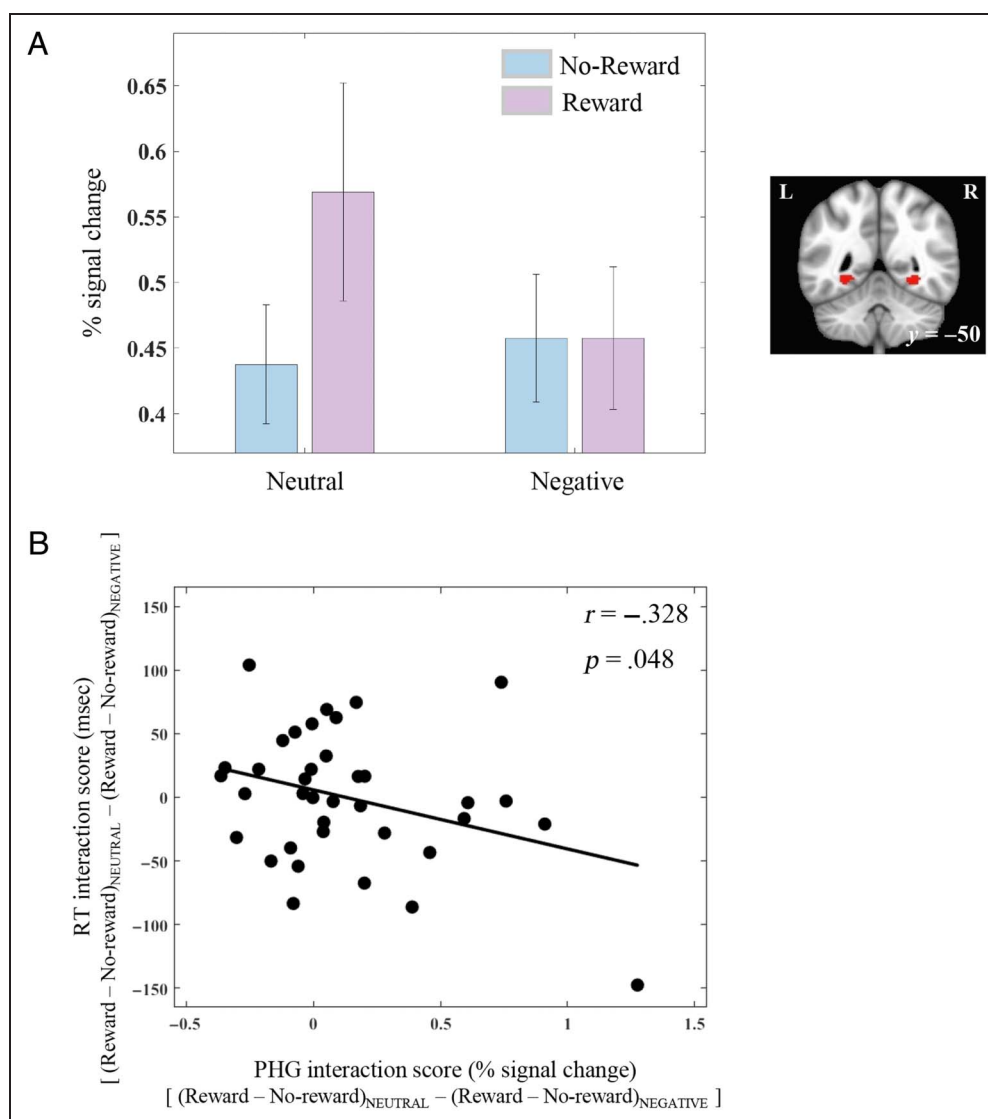


Figure 6. Cue stage ROI analysis results in (A) VS, (B) vmPFC, and (C) amygdala. For each ROI, the bar plot shows the average group-level estimates of the four experimental conditions of interest during the cue stage. The error bars represent within-subject error bars based on O'Brien and Cousineau (2014). Of note, beta estimates for cue stage regressors in VS and vmPFC were negative because of the additional task stage regressors in the same GLM. As the task-evoked responses (see Supplemental Figure S2) were higher compared to the cue-evoked responses in these regions, the beta estimates for the cue stage regressors were negative relative to an "arbitrary baseline," which is roughly equivalent to the mean of the fMRI time series that might have been augmented because of the larger task-evoked responses. L = left; R = right.

Figure 7. (A) Task stage ROI analysis results in PHG. The bar plot shows the average group-level estimates of the four experimental conditions of interest during the task stage. The error bars represent within-subject error bars based on O'Brien and Cousineau (2014). (B) Brain-behavior correlation: A negative linear relationship was observed between PHG interaction scores and behavioral RT interaction scores. L = left; R = right.



interaction was driven by higher responses in reward (relative to no-reward) trials during the neutral, $t(36) = 2.611$, $p = .013$, Cohen's $d = 0.429$, $BF_{01} = 0.30$, but not during the negative, $t(36) = 0.0004$, $p = 1.0$, Cohen's $d = 7.235e-5$, $BF_{01} = 5.66$, emotion condition. Overall, the observed two-way interaction in PHG indicated that signaling reward prospect with a negative (relative to a neutral) emotional image attenuated the reward-driven enhancements in task-related processing.

Relationship between PHG Responses and Behavior

To examine the potential link between PHG responses and behavior during the house-building discrimination task, we ran an across-participants Pearson correlation analysis between the Reward $\times$ Emotion interaction scores $[(\text{Reward} - \text{No-reward})_{\text{NEUTRAL}} - (\text{Reward} - \text{No-reward})_{\text{NEGATIVE}}]$ in PHG ROI and the corresponding RT interaction scores. We detected a significant (negative)

correlation between the PHG and RT interaction scores, $r(36) = -0.33$, $p = .048$, $BF_{01} = 0.74$ (Figure 7B). Because reward (vs. no-reward) anticipation elicited greater activity in PHG, the higher value of the PHG interaction score indicates a stronger competitive interaction between negative emotion and reward anticipation. In contrast, because RTs were faster (i.e., smaller) during the reward (relative to no-reward) anticipation, the lower value of the RT interaction score indicates stronger competitive interaction. Hence, the observed negative linear relationship indicates that participants who exhibited a stronger competitive interaction between negative emotion and reward anticipation in PHG also showed a stronger competitive interaction in RT.

Relationship between Cue-stage Responses and Task-stage PHG and RT Responses

Because the only difference between the different reward-emotion conditions while performing the house-building discrimination task was the initial cues, we first examined

whether individual differences in negative emotion and reward anticipation responses during the initial cue stage explain the variability in PHG task-related processing. Specifically, we set up a multiple linear regression model to test if the reward anticipation and negative valence responses encoded in VS and vmPFC, respectively, during the initial cue stage interactively influenced the task-related processing in PHG during the subsequent task stage. So, we employed the main effect of reward in VS (cue stage), the main effect of emotion in vmPFC (cue stage), and their interaction as the regressors, and the PHG interaction score during the task stage as the dependent variable (see Methods section). The regression coefficients corresponding to the VS reward main effect alone ($b_{VS} = -0.277, p = .432, BF_{01} = 2.30$) and the vmPFC emotion main effect alone ($b_{vmPFC} = 0.052, p = .882, BF_{01} = 1.30$) were nonsignificant. Crucially, we observed a significant relationship between the VS Reward Effect $\times$ vmPFC Emotion Effect interaction regressor and the PHG interaction scores ($b_{VS*vmPFC} = -3.864, p = .0027, BF_{01} = 0.06$). To further understand this relationship, we conducted post hoc nonstatistical analyses by splitting the participants into two subgroups based on the median split of the main effect of emotion scores in vmPFC: high negative valence activity group (below the median, as negative images elicited smaller response than neutral on average and higher negative valence is associated with lower vmPFC activity; see Heller et al., 2014) and low negative valence activity group (above the median). Then, we plotted the relationship between the VS reward main effect and PHG interaction scores separately for these two subgroups of participants (Figure 8). Participants with higher negative valence-related activity exhibited a positive

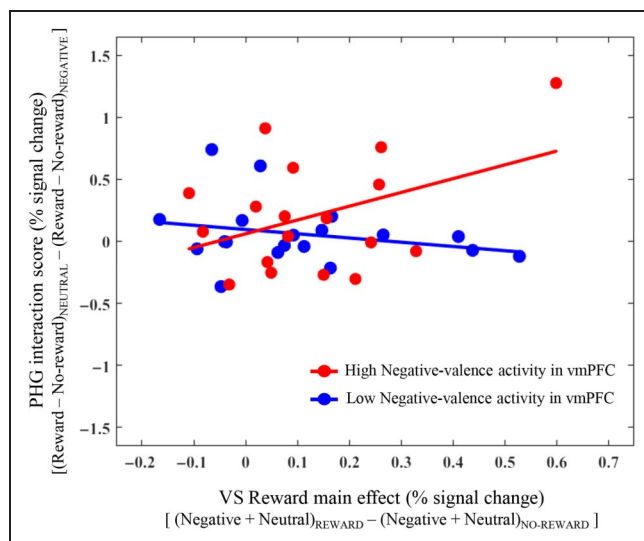


Figure 8. Moderation analysis results. In the subgroup of participants with higher negative valence-related activity in vmPFC (shown in red), the relationship between the VS reward anticipation effect and PHG interaction scores was positive. However, in the subgroup of participants with lower negative valence-related activity in vmPFC (shown in blue), the relationship was flat (or even slightly negative).

relationship between the VS reward anticipation effect and PHG interaction scores. In contrast, participants with lower negative valence-related activity had a flat (or even slightly negative) relationship. Therefore, across participants, the effects of VS reward anticipation activity on task-related processing in PHG depended on the strength of negative valence responses in vmPFC.

In a separate regression model, we additionally tested if the reward anticipation and negative valence responses during the initial cue stage impacted the behavioral RT during the subsequent task stage (see Methods section). The regression coefficient corresponding to the VS Reward main effect alone was nonsignificant ($b_{VS} = -14.038, p = .775, BF_{01} = 1.674$), and the one corresponding to the vmPFC Emotion main effect alone was marginally significant ($b_{vmPFC} = 96.629, p = .0548, BF_{01} = 0.078$). Of primary interest, we did not detect a significant relationship between the VS Reward Effect $\times$ vmPFC Emotion Effect interaction regressor and the RT interaction scores ($b_{VS*vmPFC} = 281.127, p = .1008, BF_{01} = 0.855$).

Exploratory Analyses on the Role of Gender

Finally, because some previous work has reported gender differences in negative emotion processing (Filkowski, Olsen, Duda, Wanger, & Sabatinelli, 2017; Stevens & Hamann, 2012; Bradley, Codispoti, Sabatinelli, & Lang, 2001), we conducted exploratory analyses to examine the role of gender in the amygdala and vmPFC negative emotion responses during the initial cue stage as well as the behavioral RT and PHG responses during the subsequent task stage (see Supplemental Figures S3–S4 and Supplemental Tables S3–S4). We did not detect any significant role of gender in modulating negative emotion responses in the amygdala and vmPFC during the cue stage. Similarly, no impact of gender was detected in the behavioral RT and PHG responses during the task stage. However, these null effects of gender need to be interpreted with caution, as our study was not originally designed and hence not statistically powered to examine sex differences.

DISCUSSION

In this fMRI study, using a novel variant of the MID task, we investigated the interactions between negative emotion and reward anticipation and their impact on visual perception during a subsequent house-building discrimination task. Contrary to our hypothesis, we did not detect Emotion $\times$ Reward interactions during the initial cue stage in the whole-brain voxel-wise analysis and focused a priori ROI analyses involving the VS, vmPFC, and amygdala. Notably, during the subsequent task stage, we detected a competitive interaction between negative emotion and reward anticipation in the PHG ROI. Furthermore, the Emotion $\times$ Reward interaction scores in PHG and behavioral RT were correlated across participants. Finally, the moderation analysis revealed that the relationship

between reward anticipation activity in the VS and subsequent task-related processing in PHG depended on the strength of the negative valence-related activity in vmPFC. Together, these findings indicate that neural signatures of negative emotion and reward anticipation, which were largely segregated during the cue stage, interactively modulated subsequent task-related processing, thus potentially influencing behavior.

In our paradigm, unlike in most previous MID task studies, the negative versus neutral valence of an emotional scene stimulus served as a cue to signal a performance-contingent reward (or no-reward) opportunity. The faster RT responses during reward (relative to no-reward) prospect trials indicate that participants evaluated the valence of the cue image to ascertain whether a bonus reward was at stake and heightened their performance accordingly, thus confirming the effectiveness of our motivation manipulation. However, the lack of Emotion $\times$ Reward interaction in the RT data suggests that despite being goal relevant, the effects of briefly presented negative (relative to neutral) cue images might not have been sustained during the entire delay period to impact the behavioral signatures of reward motivation during the subsequent task stage, which was further vindicated by the absence of the main effect of emotion in the RT data. Furthermore, in the subjective motivation ratings, participants reported being more motivated when the cue image signaled a reward (relative to no-reward) opportunity and less motivated when the cue image was of negative (relative to neutral) emotion. Nevertheless, as with the RT data, Emotion $\times$ Reward interaction was not detected.

In the brain, during the initial cue stage, the reward (vs. no-reward) prospect elicited higher fMRI activity in several brain regions, including the dorsal striatum, VS, thalamus, and ACC (Chen et al., 2022; Oldham et al., 2018), confirming that participants effectively employed the instructed valence–reward mapping. The central aim of this study was to investigate how negative emotion impacts neural signatures of reward anticipation. Because of incompatibility along the common value/valence dimension, we expected that reward anticipation would be attenuated when signaled by a negative image. Contrary to our hypothesis, we did not detect any evidence for the modulation of reward anticipation responses by concurrent negative emotion in the VS. In addition, because reward and emotion manipulations were intertwined in our paradigm, we probed the impact of concurrent reward anticipation on the signatures of negative emotion. In the vmPFC, negative (compared to neutral) images elicited lower activity, consistent with the past work that implicated vmPFC in encoding emotional valence (Lindquist et al., 2016; Chikazoe, Lee, Kriegeskorte, & Anderson, 2014; Heller et al., 2014; Winecoff et al., 2013). In contrast, negative (vs. neutral) images elicited greater activity in the amygdala, in line with the past studies that reported the amygdala's involvement in coding emotional arousal (Lin et al., 2020; Sabatinelli et al., 2005; Phan et al., 2003;

Adolphs, Russell, & Tranel, 1999; but see Anders et al., 2008). Although some fMRI meta-analytic findings reported the engagement of the amygdala during reward anticipation (Chen et al., 2022; Oldham et al., 2018), we did not detect the signatures of reward anticipation. In our paradigm, where emotion and reward manipulations were fused in a single cue stimulus, the amygdala appears to be relatively more sensitive to emotion than reward information. Furthermore, of primary interest, the neural signatures of negative emotion in the vmPFC and amygdala were not impacted by the concurrent reward (vs. no-reward) anticipation. Overall, the pattern of cue-evoked responses indicates that the reward anticipation and negative emotion were largely encoded independently, with a lack of evidence for interaction. However, the whole-brain voxel-wise conjunction analysis revealed additive effects of negative (vs. neutral) emotion and reward (vs. no-reward) anticipation in the rIPS, a key node of the top-down attentional network (Corbetta & Shulman, 2002). Interestingly, negative image signaling reward prospect elicited the highest response in rIPS, suggesting that both negative emotion and reward anticipation enhanced attention independently (Pessoa & Engelmann, 2010; Vuilleumier & Huang, 2009).

As noted in the Introduction, a couple of past fMRI studies that examined the impact of negative emotion on reward anticipation/motivation signatures also reported null interactions (Park et al., 2019; Wittmann et al., 2008). As the neural signatures of emotion were not detected in these studies, the lack of evidence for interaction could be a consequence of the relatively weak emotional manipulations. In the current study, despite finding strong negative emotion effects in vmPFC and amygdala, reward anticipation-related activity in the VS and several other regions was not modulated by concurrent negative emotion (see also Gieske & Sommer, 2023). Yet, it is plausible that more potent affect manipulations, such as a negative mood or stress, could modulate the neural signatures of anticipatory reward processing (Yi, Chen, Yan, Kohn, & Wu, 2023; Ironside, Kumar, Kang, & Pizzagalli, 2018; Young & Nusslock, 2016).

Notably, during the task stage entailing a house-building discrimination task, we detected a competitive Emotion $\times$ Reward interaction in the PHG ROI, a region centrally involved in the perception of visual scenes and places (Epstein et al., 1999; Aguirre et al., 1998). Specifically, reward (relative to no-reward) anticipation led to enhanced task-related processing when signaled by neutral images, which is consistent with several lines of past work (Takagaki & Krug, 2020; Rothkirch & Sterzer, 2016; Pessoa, 2015), potentially contributing to enhanced behavioral performance in perceptual tasks (Antono et al., 2022; Qin, Xue, Chen, & Zhang, 2020; Engelmann et al., 2009). In contrast, such reward-driven enhancements were diminished in the case of negative cue images, implying that negative emotion counteracted the beneficial effects of reward anticipation on task-related processing. Interestingly, the Emotion $\times$ Reward interaction scores

in PHG and RT were correlated across participants, albeit at a modest level, signifying a potential link between the interactive effects of negative emotion and reward anticipation on task-related processing and behavioral performance. Of note, the PHG–RT interaction scores correlation was detected despite the absence of Emotion $\times$ Reward interaction in the RT data. This is viable because correlation analysis capitalized on the sizeable between-subject variability in the RT interaction scores, which might have worked against detecting an interaction in the within-subject ANOVA (see Yarkoni & Braver, 2010, for further conceptual clarification). The observed interaction pattern in PHG, along with the PHG–RT correlation, suggests that negative emotion and reward anticipation modulated task-related processing in a competitive manner, potentially influencing behavior. Importantly, the competitive interaction pattern observed in PHG appears to be distinct from the attentional enhancement effects of different emotion–reward combination cues. For instance, the negative image signaling reward prospect elicited the highest activity in the rIPS presumably based on the overall attentional salience, but not in PHG during subsequent task-related processing.

In our paradigm, reward anticipation and negative emotion were manipulated orthogonally during the initial cue stage, but their interactive effects were observed only during the subsequent task stage. So, we conducted a moderation analysis to examine how individual differences in VS reward anticipation and vmPFC negative valence responses during the cue stage influenced the competitive interaction in PHG during the task stage. The analysis revealed that the relationship between VS reward anticipation and PHG competitive interaction depended on the strength of the negative valence responses in vmPFC. Specifically, in the subgroup of participants who exhibited higher negative valence-related activity in vmPFC, a post hoc nonstatistical analysis revealed a positive slope between VS reward anticipation and PHG interaction scores. This positive relationship indicates that the competitive interaction in PHG was higher in individuals with higher VS reward anticipation responses, suggesting a stronger counteracting effect of negative emotion on reward anticipation-driven task-related processing. In contrast, in the subgroup of participants who exhibited lower negative valence-related activity in vmPFC, the association between VS reward anticipation and PHG interaction scores was flat (or even slightly negative), suggesting a weaker impact of negative emotion on reward anticipation-driven task-related processing. These findings indicate that negative valence-related vmPFC activity moderated the relationship between VS reward anticipation and PHG competitive interaction. The moderation analysis results could also be interpreted in a complementary manner, that the relationship between vmPFC negative valence and PHG competitive interaction depended on the VS reward anticipation. However, we preferred the earlier interpretation as we set out to examine the

impact of negative emotion on reward anticipation in this study, and some fMRI lesion work has reported a causal influence of vmPFC on reward anticipation-related activity in VS (Pujara, Philippi, Motzkin, Baskaya, & Koenigs, 2016).

At a broader level, segregated processing of negative emotion in the amygdala (and vmPFC) and reward anticipation in VS during the initial cue stage followed by their competitive interaction during the subsequent task-related processing in PHG is similar with findings from several decision-making studies that reported initial parallel processing of reward and threat information in subcortical regions followed by their integration in cortical regions such as ACC or vmPFC (Livermore et al., 2021; Aupperle, Melrose, Francisco, Paulus, & Stein, 2014; Park, Kahnt, Rieskamp, & Heekeren, 2011; Basten, Biele, Heekeren, & Fiebach, 2010; Talmi, Dayan, Kiebel, Frith, & Dolan, 2009). However, unlike in the decision-making paradigms where interactive effects of reward and threat were frequently observed in the behavioral choices involving approach–avoidance decisions, we did not detect a Reward $\times$ Emotion interaction in our behavioral RT data (see also Gieske & Sommer, 2023). This discrepancy could be because decision-making paradigms require integration of costs and benefits associated with each option to determine an optimal choice. However, in reward motivation paradigms, which do not involve a choice component and hence no explicit value comparisons, the impact of reward–emotion integration on subsequent behavioral task performance could be weaker, especially when emotion is manipulated transiently.

Although the current study focused on the interactions between negative emotion and reward motivation during a perceptual task, some past work has examined the impact of negative emotion on reward motivation during executive function tasks (Yeung et al., 2024; Choi et al., 2015). These studies reported the counteracting effect of task-irrelevant aversive information on reward-driven behavioral enhancements. For instance, Yeung and colleagues recently reported that negative distractors nullified the effects of reward motivation during response inhibition and task-switching paradigms (Yeung et al., 2024). The fMRI results from the present study complement and extend these behavioral findings by revealing neural signatures of the counteracting effect of goal-relevant negative emotion on reward-driven enhancements during perception.

In conclusion, using a novel paradigm where a negative or neutral valenced image signaled the prospect of reward, our study demonstrated the interactive effects of reward anticipation and negative emotion on subsequent task-related processing. These findings refine our understanding of how motivation and emotion co-jointly impact visual perception to sculpt our behavior.

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Corresponding author: Srikanth Padmala, Centre for Neuroscience, Indian Institute of Science, Bangalore, KA, India 560012, e-mail: spadmala@iisc.ac.in.

Data Availability Statement

The data and codes supporting the findings of this study will be made available by the authors upon reasonable request. Supplemental Material can be accessed on this article's homepage: <https://doi.org/10.1162/JOCN.a.89>.

Author Contributions

Sagarika Jaiswal: Conceptualization; Data curation; Formal analysis; Investigation; Methodology; Resources; Software; Validation; Visualization; Writing—Original draft; Writing—Review & editing. Lakshman N. C. Chakravarthula: Conceptualization; Formal analysis; Methodology; Writing—Original draft; Writing—Review & editing. Srikanth Padmala: Conceptualization; Funding acquisition; Project administration; Supervision; Validation; Writing—Original draft; Writing—Review & editing.

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Diversity in Citation Practices

Retrospective analysis of the citations in every article published in this journal from 2010 to 2021 reveals a persistent pattern of gender imbalance: Although the proportions of authorship teams (categorized by estimated gender identification of first author/last author) publishing in the *Journal of Cognitive Neuroscience (JoCN)* during this period were $M(an)/M = .407$, $W(oman)/M = .32$, $M/W = .115$, and $W/W = .159$, the comparable proportions for the articles that these authorship teams cited were $M/M = .549$, $W/M = .257$, $M/W = .109$, and $W/W = .085$ (Postle and Fulvio, *JoCN*, 34:1, pp. 1–3). Consequently, *JoCN* encourages all authors to consider gender balance explicitly when selecting which articles to cite and gives them the opportunity to report their article's gender citation balance.

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